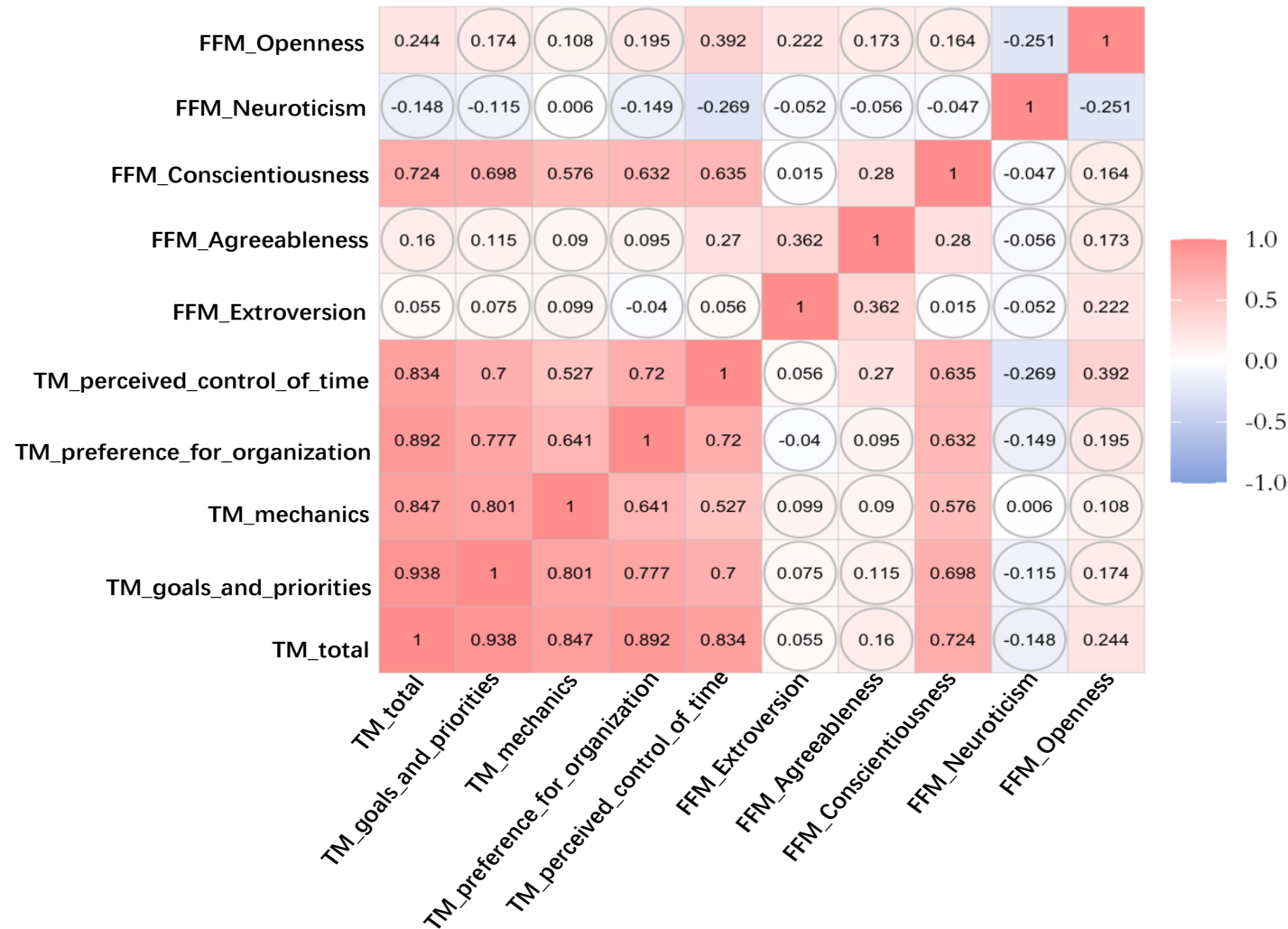


Correlation Analysis between Personality Traits and Time Management Skills – A Comprehensive View



Conscientiousness showed significant positive correlations with goal-setting ($r=0.698$), mechanics ($r=0.576$), organization preference ($r=0.632$), time control ($r=0.635$), and overall time management ($r=0.724$, all $P<0.05$). Openness was positively correlated with time control ($r=0.392$) and time management ($r=0.244$, both $P<0.05$). Agreeableness showed a positive correlation with time control ($r=0.27$, $P<0.05$), while neuroticism had a negative correlation ($r=-0.269$, $P<0.05$).

Extraversion, agreeableness, and neuroticism had slight, non-significant relationships with overall time management (all $P>0.05$). Conscientiousness and openness were significantly associated with stronger time management (both $P<0.05$)."

Note: The gray circles in the graph indicate that there is no statistically significant difference in the correlation between these two variables.

Multiple Linear Regression Analysis of Personality Traits and Time Management Skills

Multiple Linear Regression Model				
	Unstandardized B	Std. error	t-value	p-value
(Intercept)	16.508	7.345	2.248	0.027
Extroversion	0.174	0.299	0.584	0.561
Agreeableness	-0.439	0.426	-1.03	0.306
Conscientiousness	2.912	0.304	9.571	0.000
Neuroticism	-0.336	0.277	-1.21	0.230
Openness	0.507	0.366	1.385	0.170

A multiple linear regression model was established with overall time management scores as the dependent variable and the Big Five personality factors (extraversion, agreeableness, conscientiousness, neuroticism, openness) as independent variables.

Conscientiousness is the primary factor with a significantly positive impact on overall time management scores, while other personality factors do not have a statistically significant influence on overall time management scores.

Multiple Linear Regression Analysis of Personality Traits and Time Management Skills

Table 4: Adjusted Multiple Linear Regression Model

	Unstandardized B	Std. error	t-value	p-value
(Intercept)	5.841	7.547	0.774	0.441
Extroversion	0.164	0.292	0.559	0.577
Agreeableness	-0.156	0.425	-0.367	0.715
Conscientiousness	3.06	0.297	10.292	<0.001
Neuroticism	-0.177	0.273	-0.646	0.52
Openness	0.549	0.352	1.559	0.123
Gender				
Male		Ref		
Female	-0.665	1.659	-0.401	0.69
Academic grade level				
Tenth grade		Ref		
Eleventh grade	5.205	2.051	2.538	0.013
Twelfth grade	10.257	2.844	3.607	0.001
Type of school				
Regular public high school		Ref		
International school*	1.67	1.806	0.925	0.358
Daily usage of electronic devices				
More than 0.5h		Ref		
Less than 0.5h	1.653	2.365	0.699	0.487
Not use at all	2.324	2.649	0.877	0.383

An adjusted multiple linear regression is established to incorporate socio-demographic factors (gender, grade level, school type, and electrodevice usage) into the aforementioned regression model. The adjusted model results (Table 4) still indicate *conscientiousness* as a significant factor influencing overall time management scores, while *grade level* is another one. Other personality factors and socio-demographic factors do not have a statistically significant impact on overall time management scores.

Mediating Effect Analysis: Openness – Perceived Time Control Ability – Academic Stress

Table 6: Linear Regression Model Results for Mediation Analysis with Openness Serving as Independent Variable

	(1) Academic stress	(2) Perceived time control	(3) Academic stress
(Intercept)	13.968 ***	13.366 ***	13.968 ***
Openness	-0.387 *	0.501 ***	-0.254
Perceived time control	/	/	-0.265 *
R ²	0.068	0.154	0.112
Adjusted R ²	0.058	0.144	0.092

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 7: Effect Results for Mediation Analysis with Openness Serving as Independent Variable

	Effect	Standard error	p-value	Bootstrap 95% CI
Indirect effect (ab)	-0.133	0.077	0.086	[-0.299, -0.004]
Direct effect (c')	-0.254	0.185	0.170	[-0.621, 0.097]
Total effect (c)	-0.387	0.163	0.017*	[-0.713, -0.074]

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.

To examine the mediating role of perceived time control (MV), academic stress was treated as the dependent variable (DV) and openness as the independent variable (IV). Regression analyses were conducted for three paths: IV → mediator (Path a), mediator → DV (Path b), and IV → DV (total effect, Path c).

Bootstrapping with 1000 iterations estimated the 95% confidence intervals (CIs) for total and indirect effects. The CI for the **total effect** (c) ranged from -0.713 to -0.074, and the CI for the **indirect effect** (ab) ranged from -0.299 to -0.004, both excluding zero, indicating significant effects. However, the **direct effect** was not significant, as the 95% CI for the direct effect (-0.621 to 0.097) includes zero.

Mediating Effect Analysis:

Neuroticism – Perceived Time Control Ability – Academic Stress

Table 8: Linear Regression Model Results for Mediation Analysis with Neuroticism Serving as Independent Variable

	(1) Academic stress	(2) Perceived time control	(3) Academic stress
(Intercept)	13.968 ***	13.366 ***	13.968 ***
Neuroticism	0.778 ***	-0.270 **	0.738 ***
Perceived time control	/	/	-0.146
R ²	0.443	0.072	0.458
Adjusted R ²	0.437	0.062	0.4436

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 9: Effect Results for Mediation Analysis with Neuroticism Serving as Independent Variable

	Effect	Standard error	p-value	Bootstrap 95% CI
Indirect effect (ab)	0.039	0.030	0.192	[-0.007, 0.110]
Direct effect (c')	0.738	0.102	<0.001 ***	[0.537, 0.931]
Total effect (c)	0.778	0.094	<0.001 ***	[0.603, 0.959]

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.

In this analysis, neuroticism replaced openness as the independent variable (IV), while academic stress remained the dependent variable (DV) and perceived time control remained as the mediating variable (MV). Regression analyses were conducted for the same three paths: IV → mediator (Path a), mediator → DV (Path b), and IV → DV (total effect, Path c).

Bootstrapping with 1000 iterations estimated the 95% confidence intervals (CIs) for total, direct, and indirect effects. The CI for the **total effect** (c) ranged from 0.603 to 0.959, and the CI for the **direct effect** (ab) ranged from 0.537 to 0.931, both excluding zero, indicating significant effects. However, the **indirect effect** was not significant, as the 95% CI for the indirect effect (-0.007 to 0.110) includes zero.